



SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

IE2101 – REQUIREMENTS ENGINEERING (2026) ASSIGNMENT 2

INTEGRATED FURNITURE SHOP MANAGEMENT SYSTEM

Project ID: RE_G08

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INTRODUCTION

The project domain of this system is the furniture retail business. Furniture shops sell different types of products such as beds, sofas, tables, cupboards, and other home or office furniture items. In many small and medium-sized furniture shops, daily operations such as managing stock, processing orders, handling payments, and organizing deliveries are still performed manually using paper records or basic spreadsheets. Because of this, it can be difficult for businesses to manage information accurately and efficiently.

The organization involved in this project is a furniture shop that sells products directly to customers. Several roles are involved in running the business. The Shop Owner is responsible for monitoring overall business performance and making strategic decisions. System Administrators manage product information, customer orders, and overall system operations. The Finance Officer handles payments, invoices, and refunds. The Delivery Manager is responsible for scheduling deliveries, assigning delivery partners, updating delivery status, and monitoring delivery progress through the system. Delivery partners physically transport furniture to customers under the coordination of the Delivery Manager. Customer Support Officers assist customers by responding to inquiries, complaints, and feedback. Suppliers provide furniture products and coordinate stock availability with the shop.

Currently, the shop faces several problems because most processes are handled manually. Stock records can sometimes be inaccurate, which may lead to selling products that are not available. Order processing and payment handling can also be slow and may contain errors. Delivery management is another challenge, as it is difficult to track orders and schedule deliveries properly without a centralized system. In addition, the Shop Owner does not have clear and real-time information about sales, transactions, and overall business performance.

To solve these issues, the Integrated Furniture Shop Management System will provide a centralized digital platform that helps manage products, orders, payments, deliveries, and customer interactions. The system will make daily operations more organized, reduce manual errors, improve operational efficiency, and provide better customer service while supporting informed decision-making through improved data visibility.

METHODOLOGY

1. Requirements Identification

- Identified the project domain as a **furniture shop management environment**.
- Analyzed existing manual workflows such as inventory recording, order handling, payment processing, delivery scheduling, and customer support handling.
- Identified major stakeholders including:
 - Shop owner
 - System Administrators
 - Finance officer
 - Delivery manager
 - Customer support officers
 - Customers
 - Suppliers
- Categorized system functionality into six major modules:
 - User & Account Management
 - Product Management
 - Shopping Cart & Order Management
 - Payment & Billing
 - Delivery Management
 - Feedback & Support

2. Requirements Elicitation

- Conducted stakeholder interviews to understand operational problems and expected system features.
- Performed observation of current manual processes such as stock updates, order recording, and delivery coordination.
- Analyzed existing business documents including:
 - Order records
 - Payment receipts

- Delivery notes
 - Inventory lists
- Conducted **team brainstorming sessions** to refine feature ideas and improve requirement completeness.
- Developed **user stories** representing system interactions from stakeholder perspectives.
- Grouped user stories into epics to organize system functionality logically.

3. Requirements Analysis

- Reviewed collected requirements to remove inconsistencies and duplication.
- Classified requirements into:
 - Functional requirements
 - Non-functional requirements
 - System constraints
- Prioritized requirements based on business importance and stakeholder needs.
- Validated requirements using acceptance criteria derived from key system operations.

4. Requirements Modelling

- Represented structural aspects of the system using:
 - UML Class Diagram
 - UML Physical Diagram
- Represented behavioral interactions using:
 - UML Sequence Diagrams
 - UML State Diagram
- Represented workflow processes using:
 - UML Activity Diagrams
 - BPMN Process Models
- Represented decision logic using:
 - Decision Tables

- Decision Trees
- Created wireframe mock-up diagrams to illustrate system interface layout.
- These models helped visualize system structure, interactions, workflows, and decision logic from multiple perspectives.

5. Requirements Documentation

- Documented requirements in a structured **Software Requirements Specification (SRS)** format.
- Maintained traceability between:
 - Stakeholder needs
 - User stories
 - Functional requirements
 - System models
- Ensured consistency between Assignment-1 requirement outputs and Assignment-2 modelling diagrams.
- Prepared diagrams and supporting explanations to improve readability and understanding of system functionality.

DISCUSSION

Difficulties Encountered During Requirements Analysis

- Faced challenges in understanding real-world furniture shop workflows due to limited access to a fully digitized furniture retail environment.
- Required making reasonable assumptions about delivery scheduling processes and inventory coordination procedures.
- Had trouble maintaining consistency between interconnected modules such as order management, payment processing, and delivery tracking.
- Selecting appropriate use cases for sequence diagrams, activity diagrams, and BPMN models required multiple revisions to ensure correct workflow representation.
- Ensuring alignment between functional requirements and UML modelling elements required careful validation across diagrams.

Limitations of the Analysis

- Some system behaviors related to external payment gateway integration were assumed because real integration environments were not available.
- Delivery route optimization and logistics planning features were simplified due to limited access to real operational delivery datasets.
- Supplier coordination workflows were analyzed at a conceptual level rather than at enterprise integration level.
- The analysis focused mainly on core business operations and did not include advanced analytics or reporting automation features.
- Certain non-functional performance metrics such as exact system scalability limits were estimated rather than experimentally validated.

Lessons Learned During Requirements Specification Process

- Learned the importance of using multiple requirement elicitation techniques such as interviews, observation, and document analysis to improve requirement accuracy.
- Understood how grouping user stories into system modules improves requirement organization and traceability.

- Gained experience in converting informal stakeholder needs into structured functional and non-functional requirements.
- Improved understanding of how UML diagrams and BPMN process models represent system behavior from different perspectives.
- Recognized the importance of maintaining consistency between requirement descriptions and system models.
- Learned how decision tables and decision trees help represent complex conditional system behaviors clearly.
- Developed practical experience in preparing structured Software Requirements Specification (SRS) documentation aligned with software engineering standards.

FUNCTIONAL REQUIREMENTS

1) User & Account Management

FR1	Register Account
Input	Name, email address, phone number, password
Processing	System validates user input, checks duplicate email existence, creates user account record, sends verification email
Output	User account created confirmation message
Definition	User details include name, email, phone number, and password

FR2	Secure Login
Input	Registered email address, password
Processing	System verifies authentication credentials against stored records

Output	User dashboard displayed upon successful login
Definition	Access granted only for valid registered users

FR3	Update Profile
Input	Updated profile information
Processing	System validates updated data and updates user profile database
Output	Profile update confirmation message
Definition	Profile includes name, phone number, and delivery address

FR4	Reset Password
Input	Registered email address
Processing	System verifies account and sends password reset link
Output	Password reset email notification
Definition	Reset link allows secure password change

FR5	Email Verification
Input	User email address

Processing	System sends verification link and confirms validation
Output	Email verification confirmation message
Definition	Verification ensures authenticity of user identity

FR6	View Account Details
Input	User request
Processing	System retrieves stored user account details and login history
Output	Account details displayed
Definition	Account details include profile information and login activity records

FR7	Logout
Input	Logout request
Processing	System terminates active session securely
Output	Logout confirmation message
Definition	Session termination prevents unauthorized access

FR8	Select Language Preference
Input	Selected language option

Processing	System updates interface language configuration
Output	Interface displayed in selected language
Definition	System supports multilingual usability

2) Product Management

FR9	Add Product
Input	Product ID, Product name, category, description, price, image
Processing	System validates and stores product record
Output	Product added confirmation message
Definition	Product details stored in inventory database

FR10	Update Product
Input	Product ID, updated product information
Processing	System modifies stored product record
Output	Product update confirmation message
Definition	Updated details replace previous product data

FR11	Delete Product
Input	Product ID

Processing	System removes product record
Output	Product deletion confirmation message
Definition	Deleted products removed from customer view

FR12	Upload Product Image
Input	Product image file
Processing	System uploads and links image to product
Output	Image upload confirmation message
Definition	Images enhance product visualization

FR13	Manage Product Categories
Input	Category creation/update/delete request
Processing	System updates category records
Output	Category update confirmation message
Definition	Categories organize inventory logically

FR14	View Product List
Input	Customer browsing request

Processing	System retrieves available products
Output	Product list displayed
Definition	Includes product name, category, price, and image

FR15	View Product Details
Input	Product ID
Processing	System retrieves detailed product information
Output	Product details displayed
Definition	Details include description and availability

FR16	Search Products
Input	Search keyword
Processing	System filters product database
Output	Search results displayed
Definition	Supports keyword-based search functionality

FR17	View Related Products
Input	Selected product ID

Processing	System retrieves similar category products
Output	Related product suggestions displayed
Definition	Supports alternative product discovery

3) Shopping Cart & Order Management

FR18	Add Item to Cart
Input	Product ID, quantity
Processing	System adds selected product to shopping cart
Output	Item added to cart confirmation message
Definition	Cart temporarily stores selected products before order placement

FR19	Remove Item from Cart
Input	Product ID
Processing	System removes selected product from cart
Output	Item removed confirmation message
Definition	Cart reflects updated product selection

FR20	Update Cart Quantity
Input	Product ID, updated quantity

Processing	System updates quantity value of selected product
Output	Cart updated confirmation message
Definition	Quantity changes affect order total calculation

FR21	Place Order
Input	Cart items, delivery details, payment method
Processing	System validates cart contents and creates order record
Output	Order confirmation message
Definition	Order record includes selected items and delivery information

FR22	Receive Order Confirmation
Input	Order details
Processing	System generates and sends order confirmation notification
Output	Order confirmation received message
Definition	Confirmation ensures successful order placement

FR23	Track Order
Input	Order ID
Processing	System retrieves order status information

Output	Order status displayed
Definition	Status reflects order lifecycle progress

FR24	View Order History
Input	User account request
Processing	System retrieves previous order records
Output	Order history displayed
Definition	History includes completed and cancelled orders

FR25	Cancel Order
Input	Order ID
Processing	System validates cancellation eligibility and cancels order
Output	Order cancellation confirmation message
Definition	Cancellation allowed before dispatch stage

FR26	Manage Orders
Input	Order ID, updated order status
Processing	Administrator updates order information
Output	Order status updated confirmation

Definition	Status updates reflect order processing progress
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4) Payment & Billing

FR27	Make Payment
Input	Card details, payment amount
Processing	System validates payment information and processes transaction securely
Output	Payment success confirmation message
Definition	Transactions processed through secure payment gateway

FR28	Save Card Details
Input	Card information
Processing	Administrator updates order information
Output	Card saved confirmation message
Definition	Stored details support faster checkout

FR29	Send Email Invoice
Input	Order ID, customer email
Processing	System generates invoice and sends email notification
Output	Invoice email sent confirmation

Definition	Invoice contains transaction summary
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FR30	View Transaction History
Input	User account request
Processing	System retrieves payment records
Output	Transaction history displayed
Definition	Includes completed payment transactions

FR31	Search Transactions
Input	Transaction ID or filter criteria
Processing	System searches transaction database
Output	Matching transaction records displayed
Definition	Supports administrative transaction monitoring

FR32	Process Refund
Input	Transaction ID
Processing	System validates refund eligibility and processes refund
Output	Refund confirmation message displayed.

Definition	Refund reverses completed transaction
FR33	View Failed Transactions
Input	Administrator request
Processing	System retrieves failed payment records
Output	Failed transactions list displayed
Definition	Supports payment issue monitoring

FR34	Log Payment Activities
Input	Payment event data
Processing	System records payment activity logs
Output	Payment log stored confirmation
Definition	Logs support auditing and fraud detection

FR35	Generate Sales Reports
Input	Monthly sales data
Processing	System analyses sales transaction records
Output	Monthly sales report generated

Definition	Reports support business performance evaluation
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5) Delivery Management

FR36	Create Delivery Record
Input	Order ID, delivery information
Processing	System creates delivery record
Output	Delivery record created confirmation
Definition	Record includes delivery schedule and assigns staff

FR37	Assign Delivery Details
Input	Delivery staff, delivery date
Processing	System assigns delivery schedule
Output	Delivery assignment confirmation message
Definition	Assignment links delivery personnel to order

FR38	View Delivery Details
Input	Order ID

Processing	System retrieves delivery information
Output	Delivery details displayed
Definition	Includes delivery address and schedule

FR39	Track Delivery
Input	Delivery ID
Processing	System retrieves delivery status
Output	Delivery status displayed
Definition	Status reflects shipment progress

FR40	Update Delivery Status
Input	Delivery ID, updated status
Processing	System updates delivery information
Output	Delivery status updated confirmation
Definition	Status reflects delivery lifecycle stage

FR41	Change Delivery Address
Input	New delivery address

Processing	System validates and updates delivery address
Output	Address change confirmation
Definition	Address updates allowed before dispatch

FR42	Monitor Deliveries
Input	Administrator request
Processing	System retrieves all delivery records
Output	Delivery monitoring dashboard displayed
Definition	Dashboard supports delivery tracking management

FR43	Reschedule Delivery
Input	New delivery date
Processing	System updates delivery schedule
Output	Delivery rescheduled confirmation
Definition	Schedule changes reflect updated delivery plan

FR44	Cancel Delivery
Input	Delivery ID
Processing	System cancels delivery record

Output	Delivery cancellation confirmation
Definition	Cancelled deliveries removed from schedule

6) Feedback & Support

FR45	Rate Service
Input	Rating value
Processing	System records customer service rating
Output	Rating submitted confirmation message
Definition	Ratings support service quality evaluation

FR46	Write Review
Input	Review text
Processing	System stores customer review
Output	Review submitted confirmation message
Definition	Reviews provide qualitative feedback

FR47	Edit/Delete Review
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Input	Review ID, update or delete request
Processing	System updates or removes review record
Output	Review updated/deleted confirmation message
Definition	Allows correction of submitted feedback

FR48	Create Support Ticket
Input	Issue description
Processing	System creates support ticket record
Output	Ticket ID generated confirmation
Definition	Ticket tracks customer issue resolution

FR49	Track Ticket Status
Input	Ticket ID
Processing	System retrieves ticket progress
Output	Ticket status displayed
Definition	Status includes open, pending, resolved stages

FR50	View Tickets
Input	Administrator request

Processing	System retrieves support ticket records
Output	Ticket list displayed
Definition	Supports ticket management operations

FR51	Reply to Tickets
Input	Ticket ID, response message
Processing	System stores administrator response
Output	Response sent confirmation message
Definition	Responses assist issue resolution

FR52	Update Ticket Status
Input	Ticket ID, status update
Processing	System updates ticket status record
Output	Ticket status updated confirmation
Definition	Status reflects issue resolution progress

FR53	Moderate Feedback
Input	Review ID

Processing	Administrator approves or removes feedback
Output	Feedback moderation confirmation message
Definition	Ensures professionalism of published reviews

FR54	Generate Feedback Reports
Input	Customer feedback data
Processing	System analyses feedback records
Output	Feedback report generated
Definition	Reports support service improvement decisions

NON – FUNCTIONAL REQUIREMENTS

NFR1	Performance
Description	System pages including product lists and order details shall load within 2 seconds under normal operating conditions.

NFR2	Security
Description	All payment transactions must use encrypted communication and secure authentication mechanisms.

NFR3	Availability
Description	The system shall be available 24 hours per day for customer access except during scheduled maintenance periods.

NFR4	Reliability
Description	System shall ensure transactional integrity using atomic database commits and backup recovery mechanisms.

NFR5	Usability
Description	New users shall be able to complete a purchase transaction within 3 minutes without training.

NFR6	Maintainability
Description	System architecture shall support modular updates without affecting existing functionality.

CONSTRAINTS

C1	Payment Gateway Constraint
Description	System must integrate with a secure third-party payment gateway supporting encrypted transaction processing

C2	Regulatory Constraint
Description	System must comply with financial transaction security and data privacy regulations.

C3	Development Methodology Constraint
Description	System development must follow Agile Scrum methodology.

C4	Platform Constraint
Description	System must integrate with inventory management and delivery management processes.

UML CLASS DIAGRAM

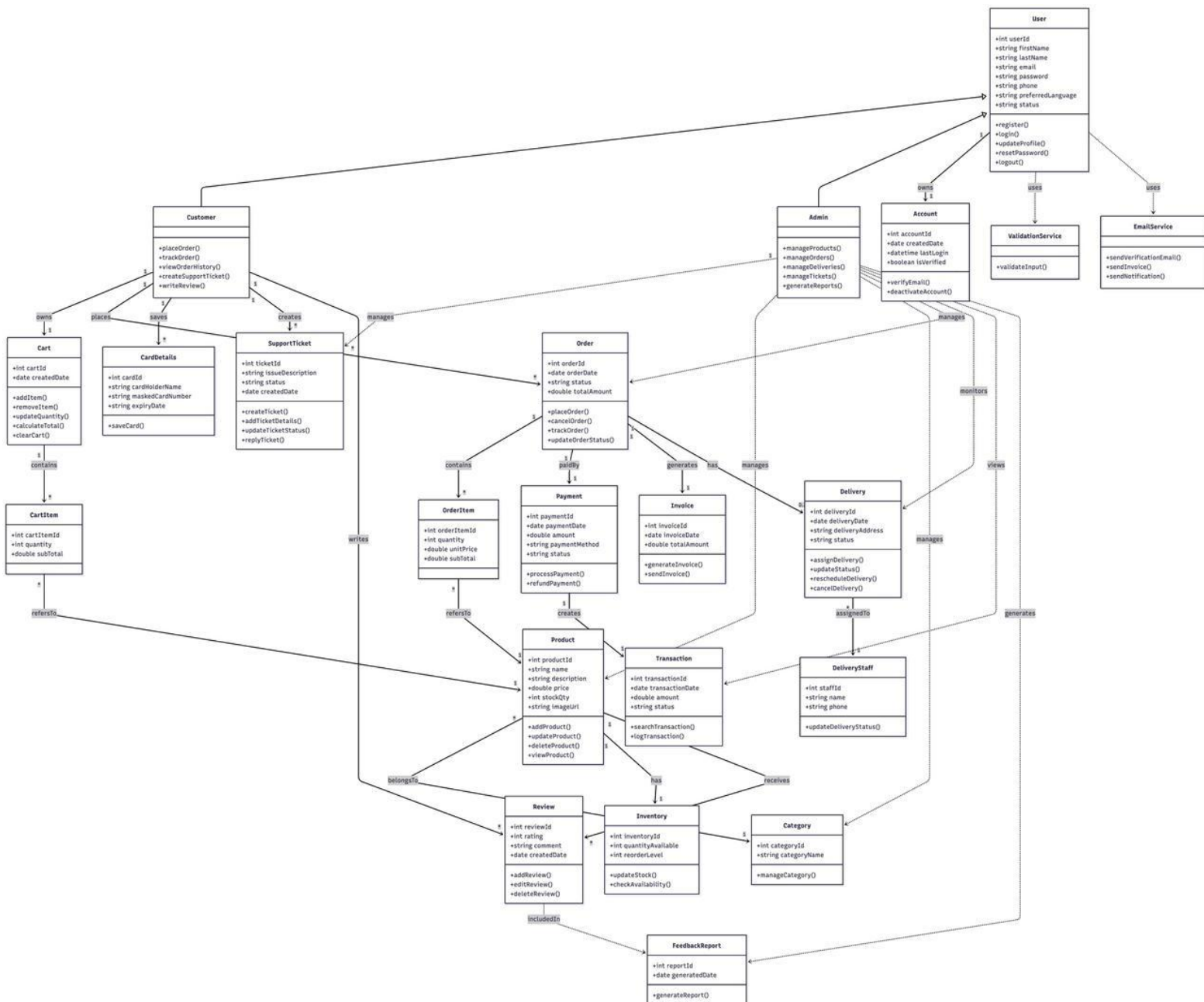
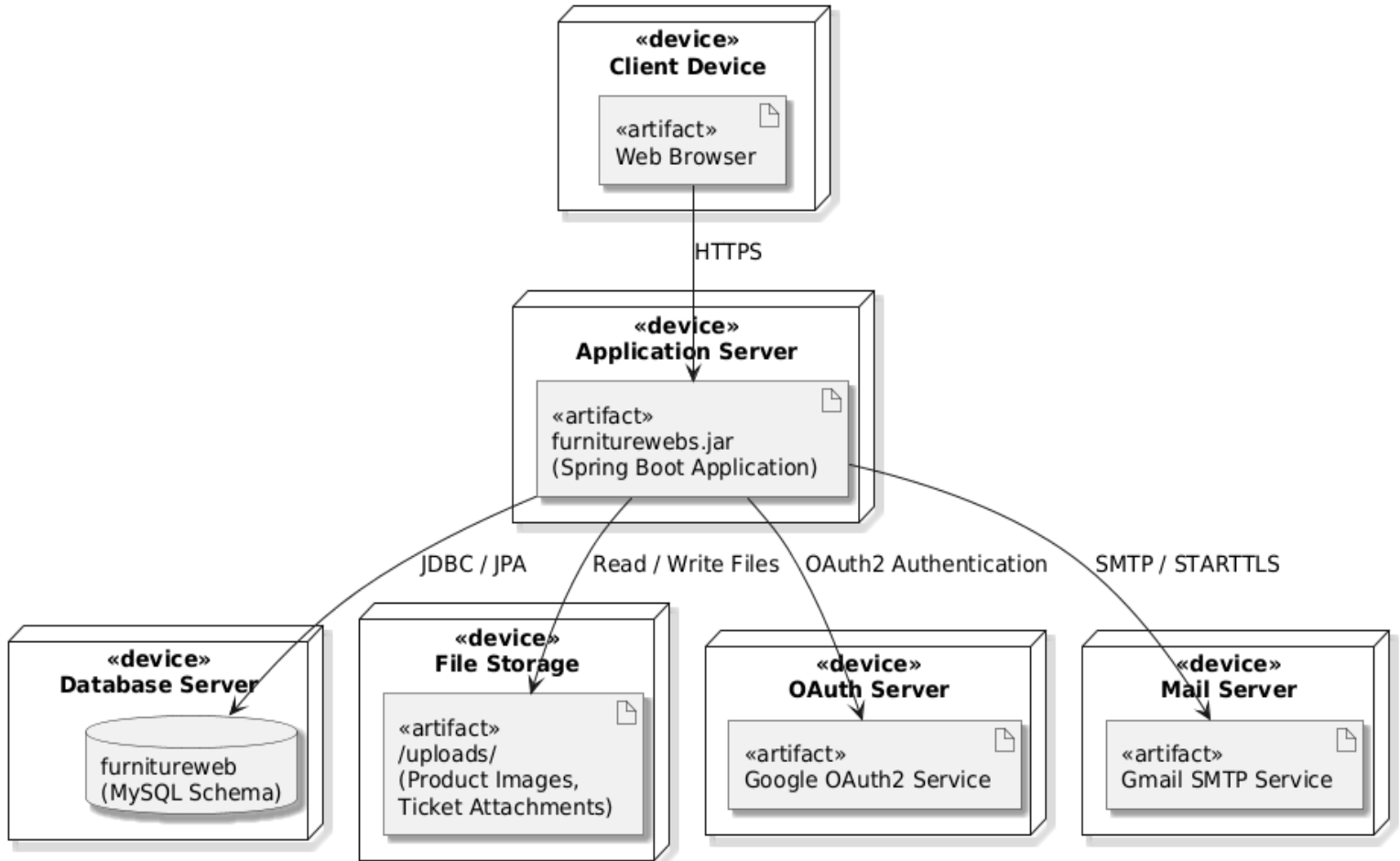


Figure 1: UML class diagram

The UML Class Diagram illustrates the main structural components of the Integrated Furniture Shop Management System, including key entities such as User, Product, Order, Payment, Delivery, and Support Ticket. It shows the relationships between classes that support inventory management, order processing, payment handling, delivery tracking, and customer support functions. This diagram represents how system components interact to support overall business operations.

UML PHYSICAL DIAGRAM

UML Deployment Diagram - FurnitureWebs



*Figure 2: UML Physical diagram
– Deployment diagram*

The UML Deployment Diagram illustrates the physical architecture of the Integrated Furniture Shop Management System and shows how system components are distributed across different hardware nodes. Users access the system through a Client Device using a web browser over secure HTTPS communication.

The Application Server hosts the main system application and handles core business operations such as order processing, product management, payment handling, and delivery coordination. It communicates with the Database Server for storing system data and uses File Storage to manage uploaded product images and attachments. External services such as Google OAuth2 support authentication, while Gmail SMTP services handle email notifications such as verification messages and invoices.

Overall, the deployment architecture ensures secure communication, efficient data management, and integration with external authentication and messaging services.

UML SEQUENCE DIAGRAMS

1) User registration

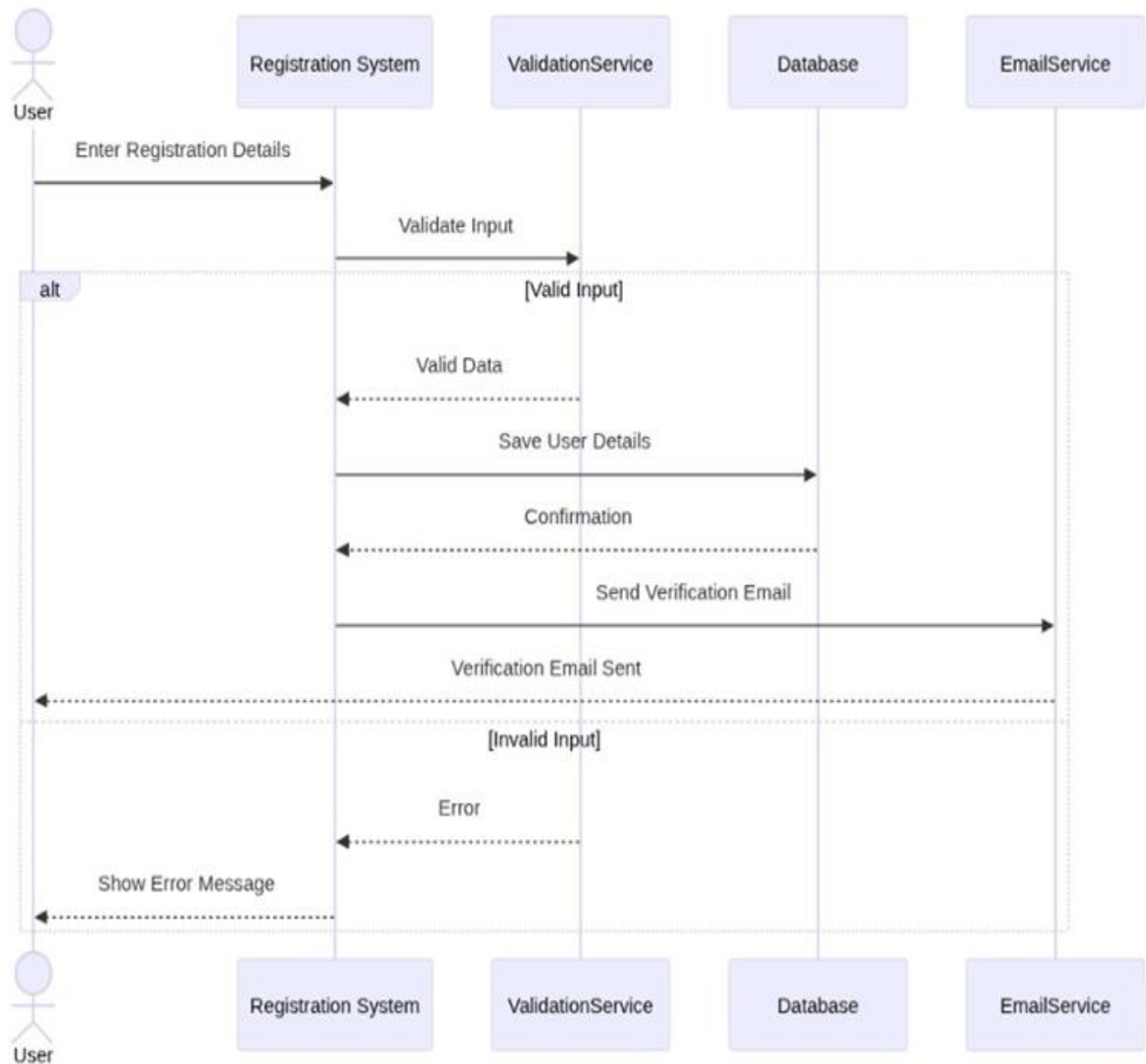


Figure 3: UML sequence diagram – user registration

This sequence diagram illustrates the interaction between the User, Registration System, ValidationService, Database, and EmailService during the account registration process. The system validates user input and stores the user details if the information is valid. After successful registration, a verification email is sent to the user. If invalid input is detected, an error message is displayed.

2) Payment management

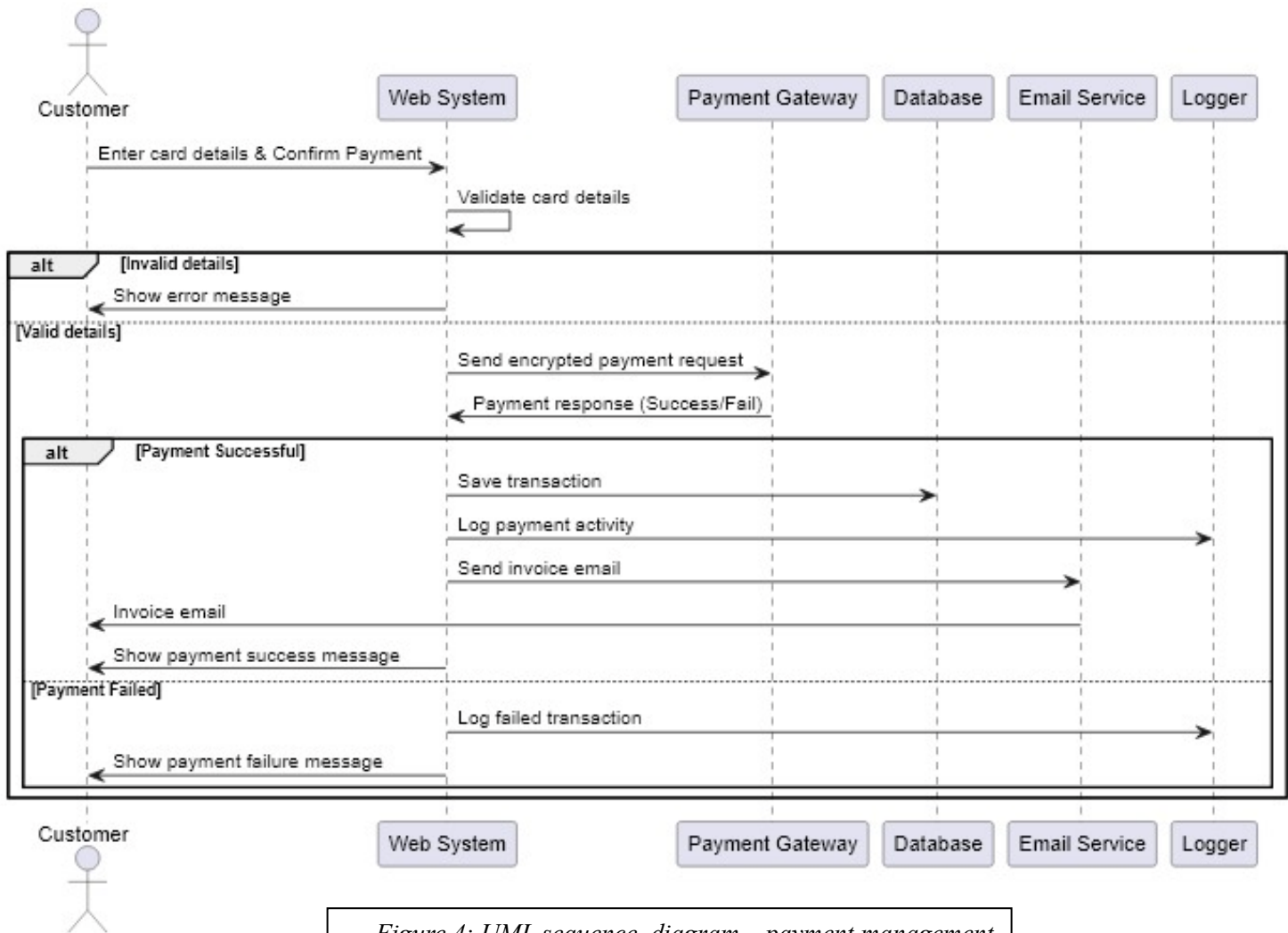


Figure 4: UML sequence diagram – payment management

This sequence diagram represents the interaction between the Customer, Web System, Payment Gateway, Database, Email Service, and Logger during payment processing. The system validates card details and sends an encrypted request to the payment gateway. If the payment is successful, the transaction is stored, payment activity is logged, and an invoice email is sent. If the payment fails, the system logs the failed transaction and displays an error message.

3) Order management

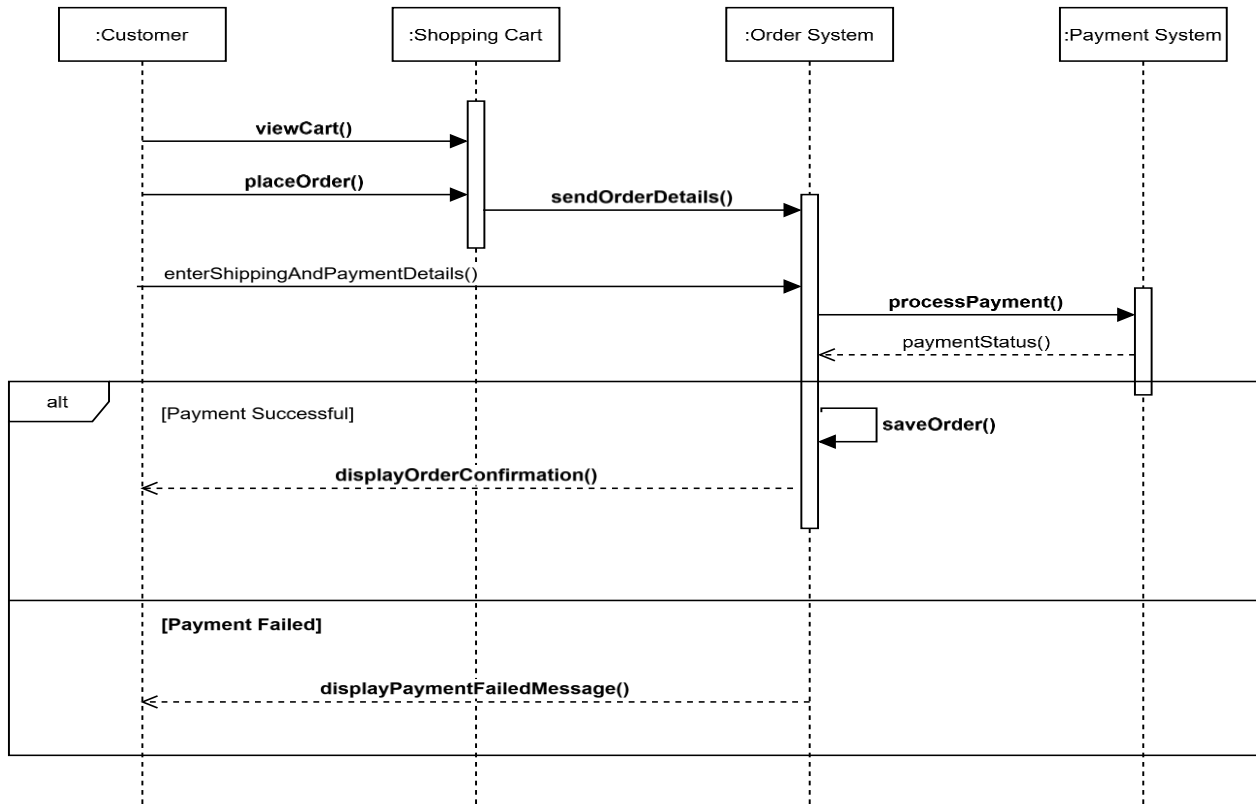


Figure 5: UML sequence diagram – order management

This sequence diagram illustrates the interaction between the Customer, Shopping Cart, Order System, and Payment System during order placement. The customer reviews the cart, enters shipping and payment details, and confirms the order. The system processes payment and saves the order if the transaction is successful. If the payment fails, the system displays a payment failure message.

4) Delivery management - IT24101357– Bandara N.H.M.W.W.D.S

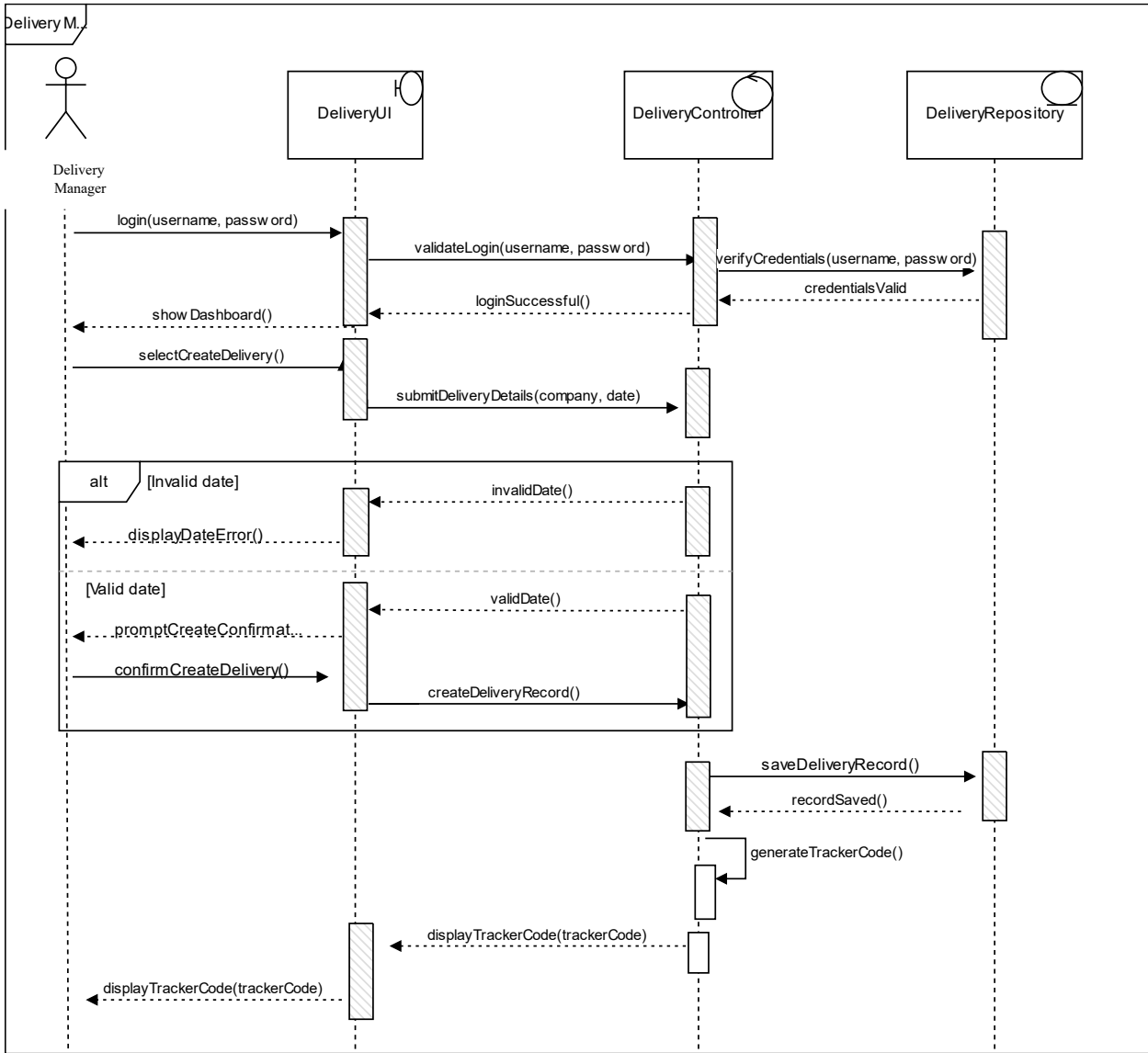


Figure 6: UML sequence diagram –Delivery management

This sequence diagram represents the interaction between the Delivery manager, DeliveryUI, DeliveryController, and DeliveryRepository during delivery record creation. The Delivery manager logs into the system and submits delivery details such as company and delivery date. The system validates the input and stores the delivery record in the database. After successful creation, a tracker code is generated and displayed to the delivery manager.

UML ACTIVITY DIAGRAMS

1) User registration

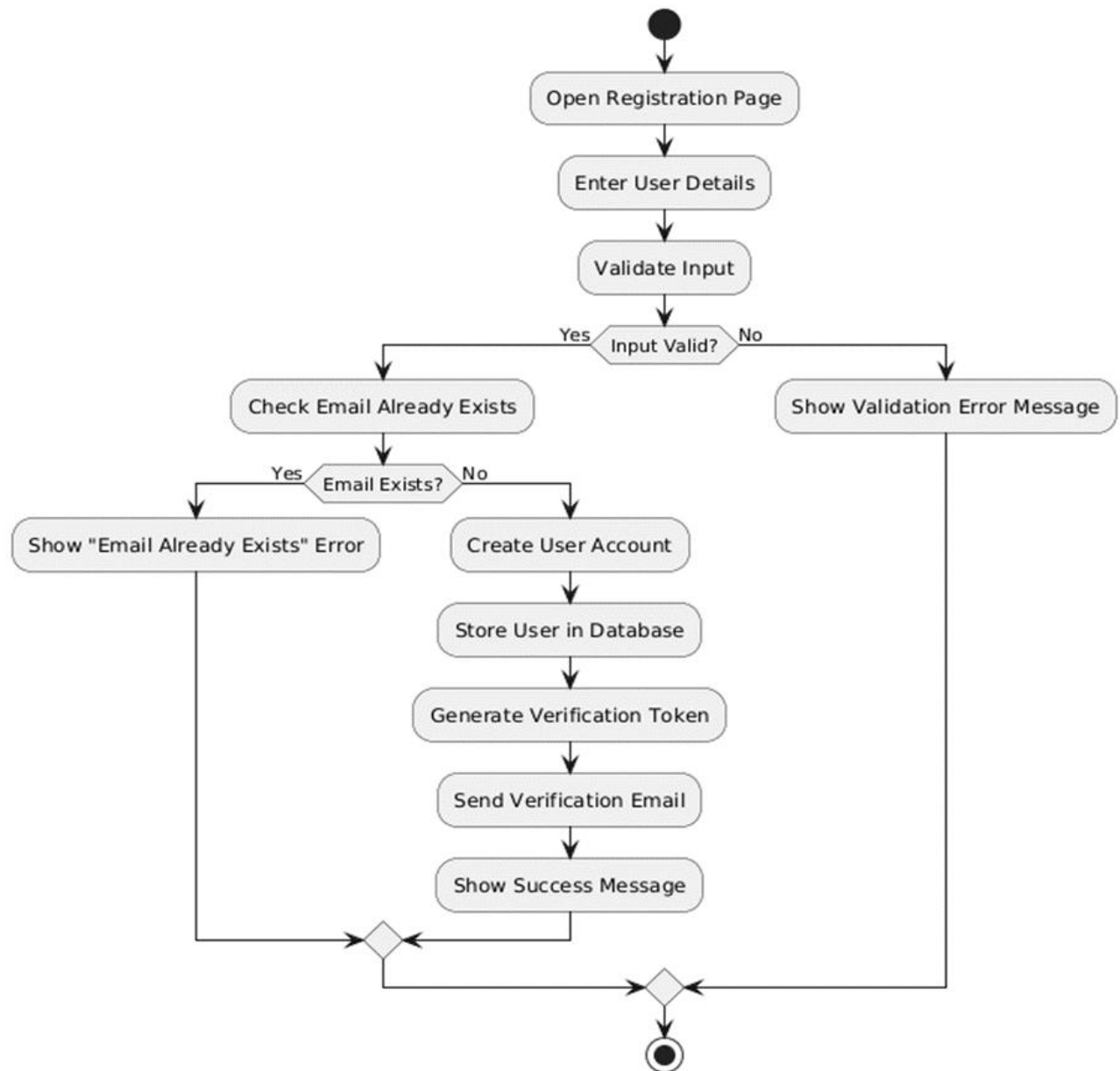


Figure 7 : Activity diagram – User registration

This activity diagram illustrates the workflow for the user registration process in the system. It shows how the system validates user input, checks whether the email already exists, and creates a new user account if the information is valid. The system then stores the user details in the database and sends a verification email. If validation fails or the email already exists, appropriate error messages are displayed to the user.

2) Payment management

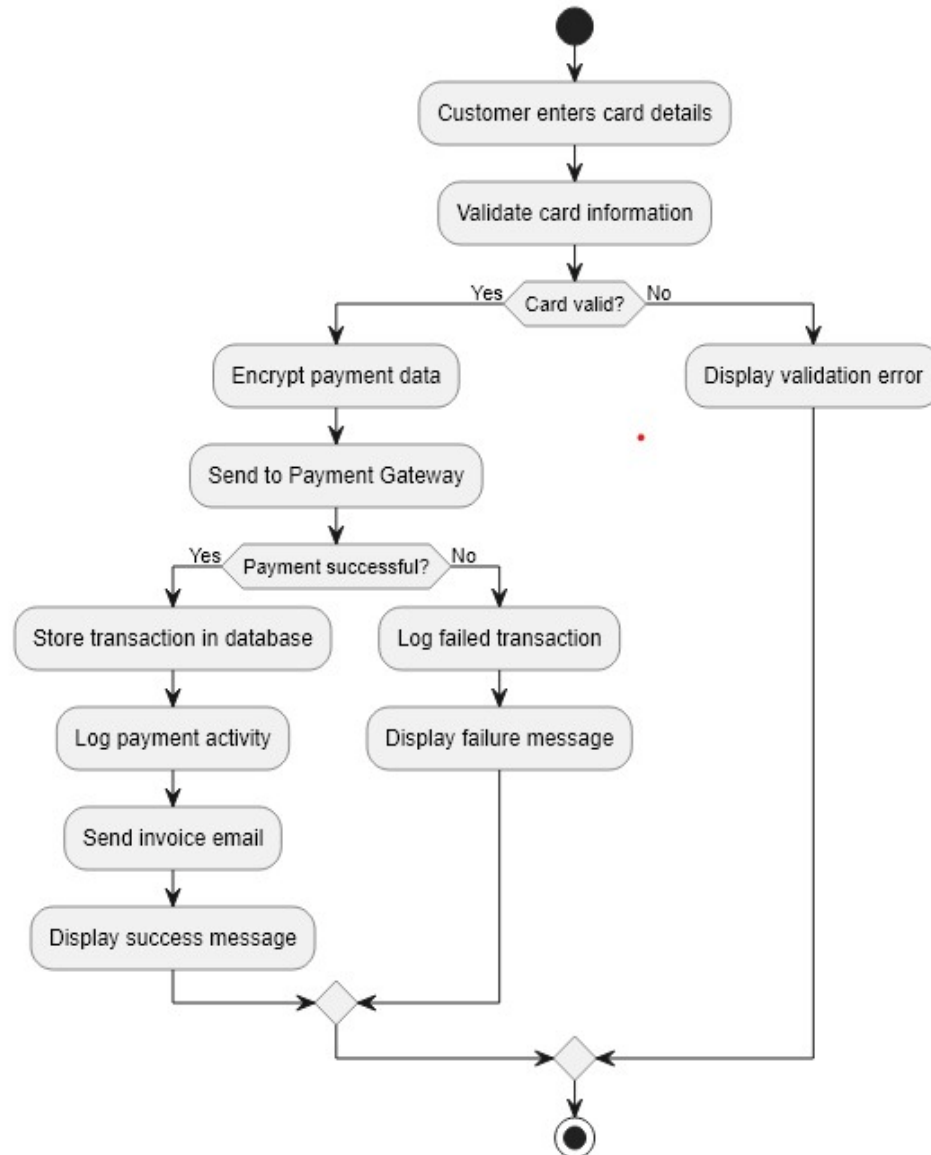


Figure 8 : Activity diagram – payment management

This activity diagram represents the payment processing workflow after the customer enters card details. The system validates the card information, encrypts payment data, and communicates with the payment gateway to process the transaction. If the payment is successful, the transaction is stored, payment activity is logged, and an invoice email is sent to the customer. If the payment fails, the system logs the failed transaction and displays an error message.

3) Order management

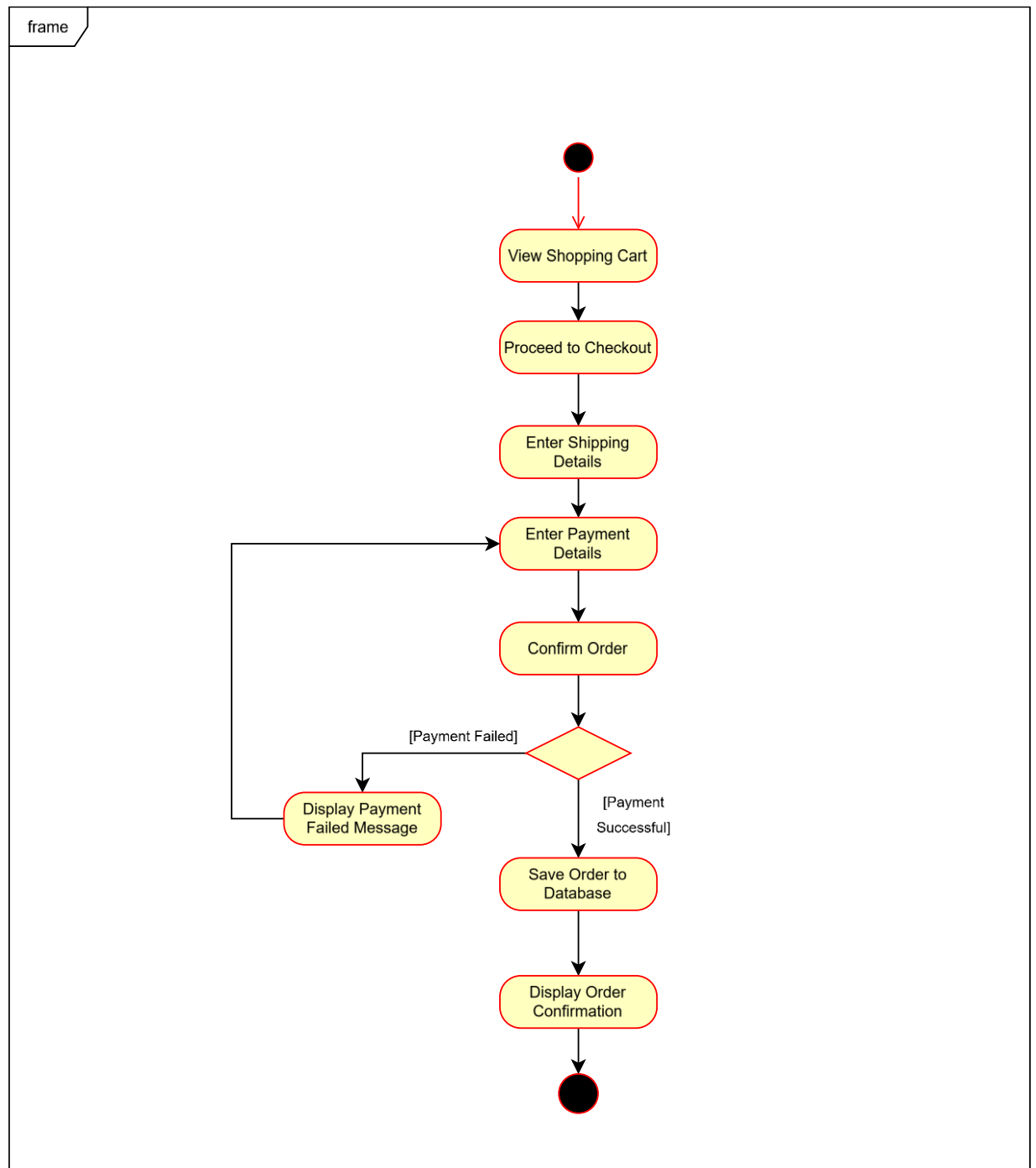


Figure 9 : Activity diagram – order management

This activity diagram illustrates the order checkout workflow followed by the customer when placing an order. It includes steps such as viewing the shopping cart, entering shipping and payment details, and confirming the order. The system verifies payment status before storing the order in the database. If payment fails, an error message is displayed; otherwise, the system confirms successful order placement.

4) Delivery management

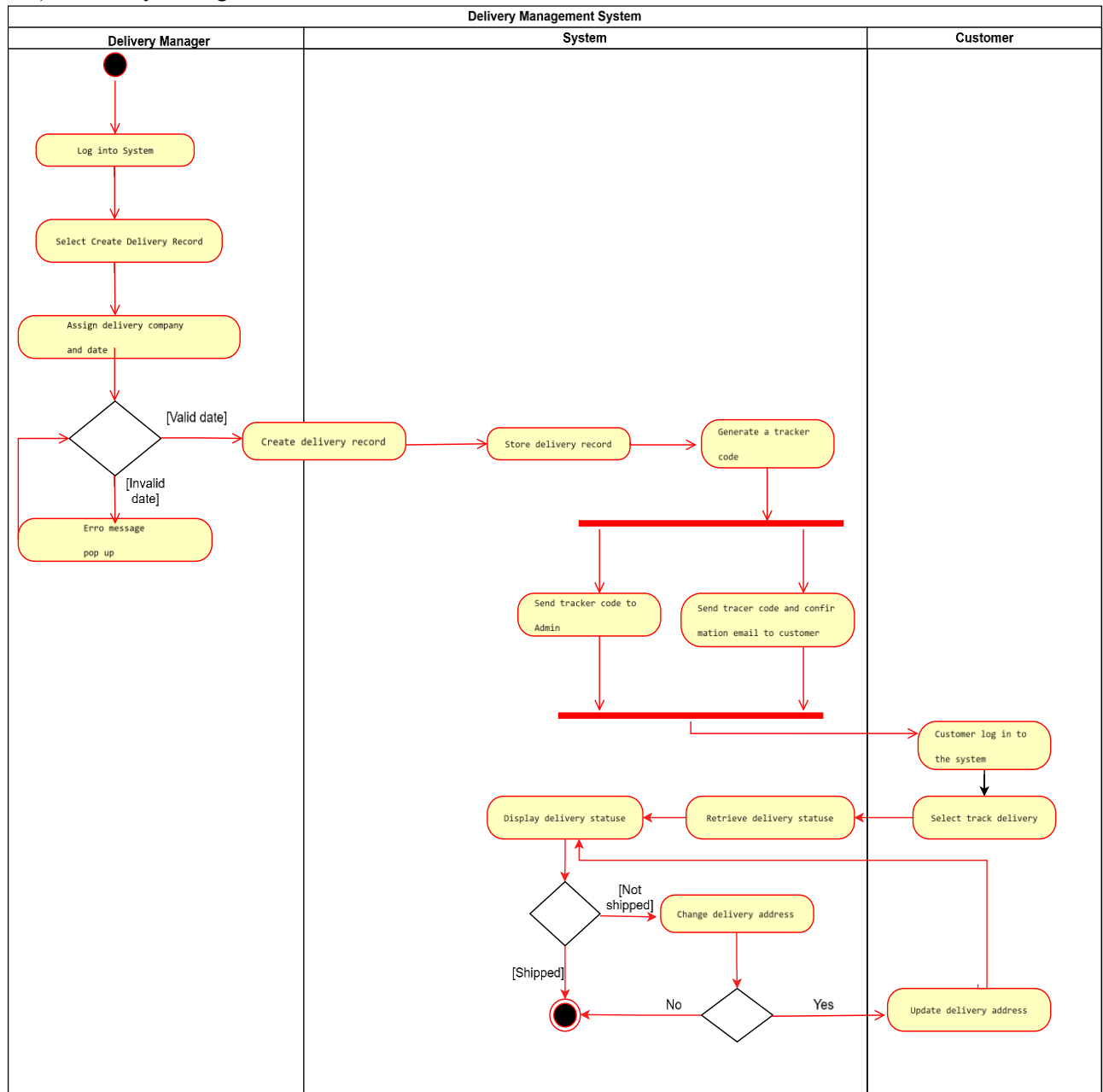


Figure 10: Activity diagram – Delivery management

This activity diagram represents the delivery management workflow involving the Delivery Manager, system, and customer. It shows how the Delivery Manager creates delivery records, assigns delivery details, and generates tracking information for customers. The customer can track delivery status and update delivery address if the order has not yet been shipped. The diagram illustrates coordination between system actors to manage delivery tracking efficiently.

UML STATE DIAGRAM

Order :

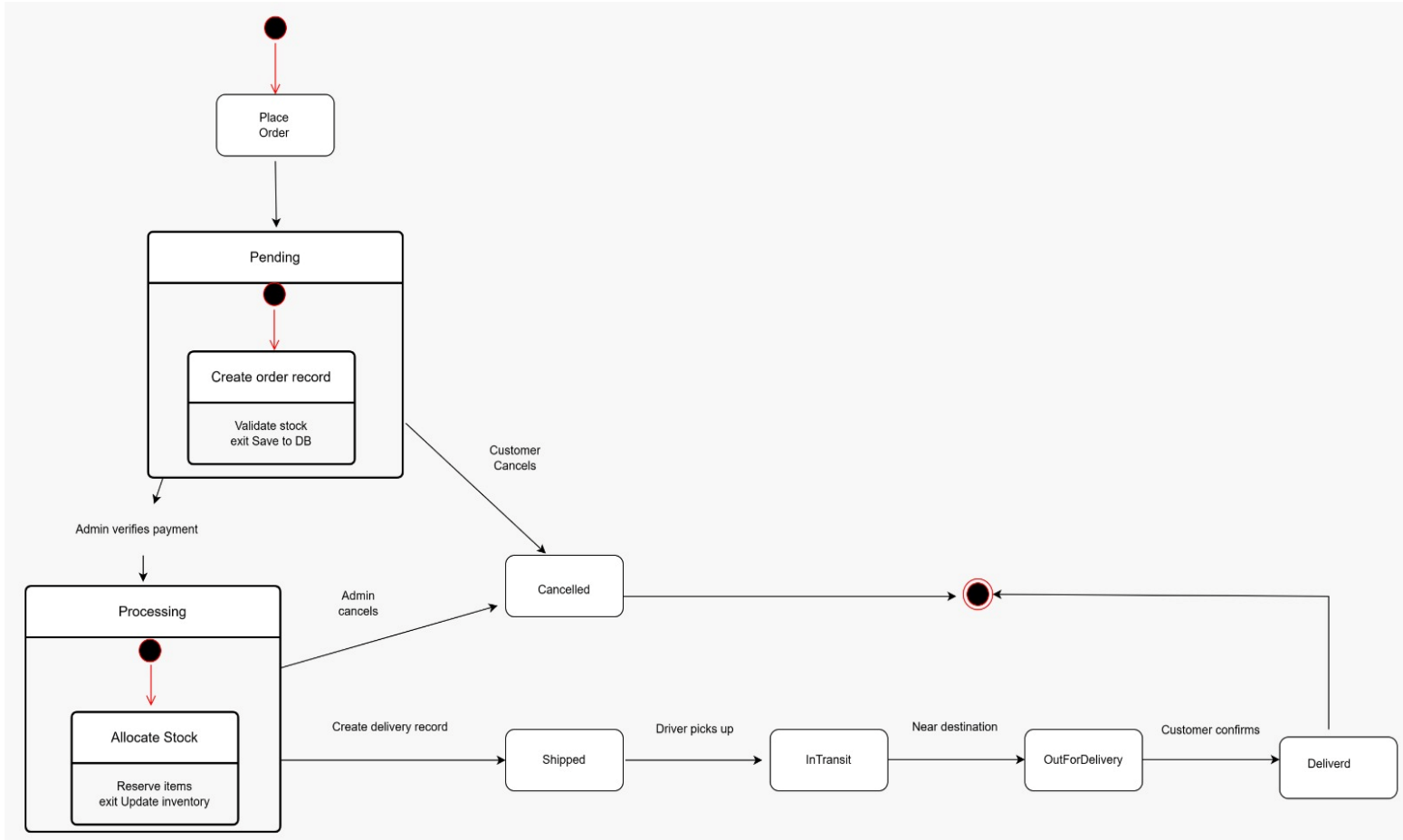


Figure 11: UML state diagram : order

This UML State Diagram illustrates the lifecycle of an order in the Integrated Furniture Shop Management System. It shows how an order transitions through different states such as Pending, Processing, Shipped, InTransit, OutForDelivery, and Delivered based on system actions and delivery progress. The diagram also represents alternative transitions such as customer or administrator cancellation before shipment. This state model helps explain how order status changes throughout the order fulfillment process.

PROCESS MODELS – BPMN DIAGRAMS

1) User registration

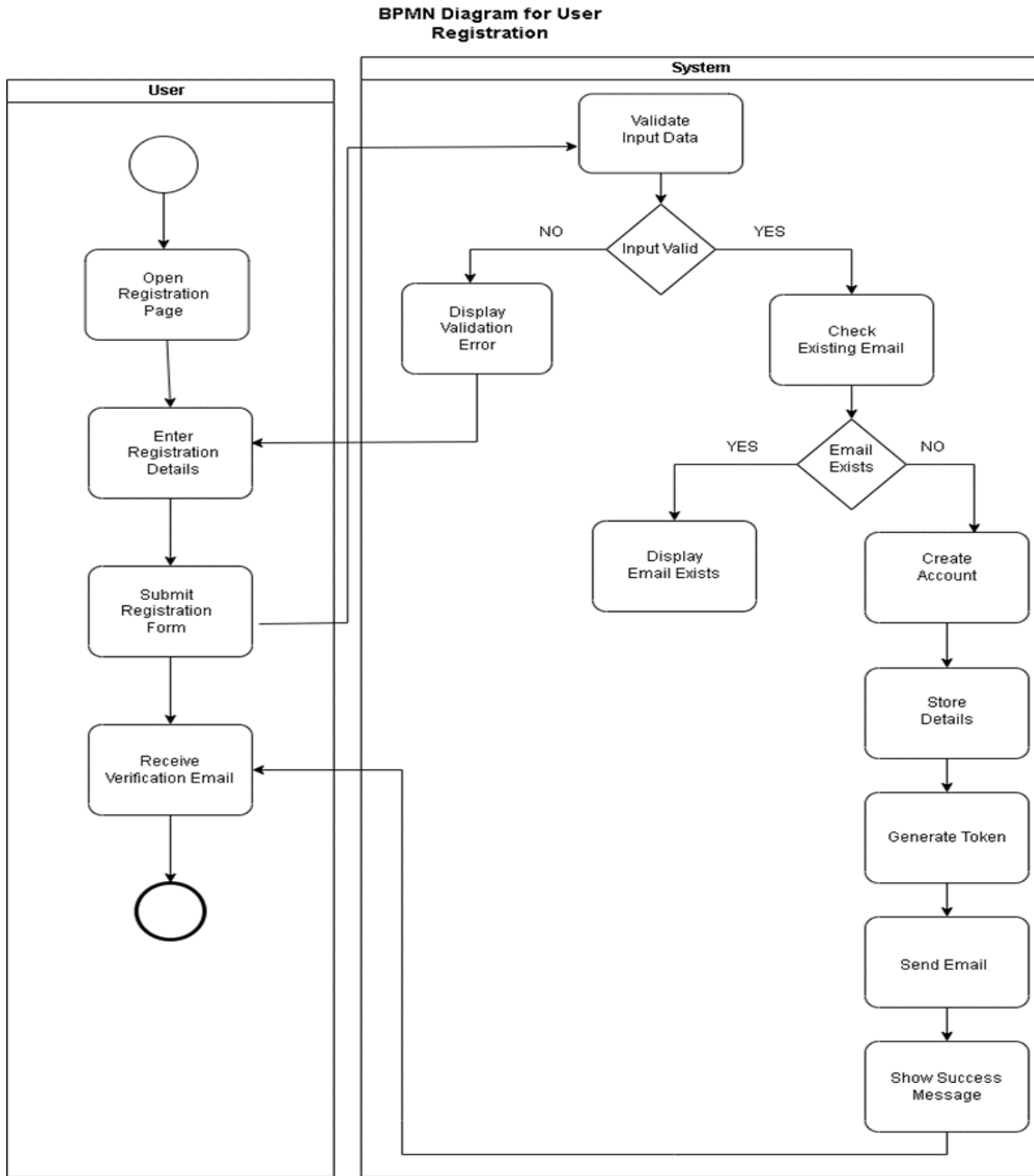


Figure 12: BPMN diagram – User registration

This BPMN diagram illustrates the business workflow of the user registration process involving the User and the System. It shows how the system validates registration details, checks for existing email addresses, and creates a new user account if the input is valid. The system then stores user information and sends a verification email to the user. Error messages are displayed if invalid input or duplicate email addresses are detected.

2) Payment management

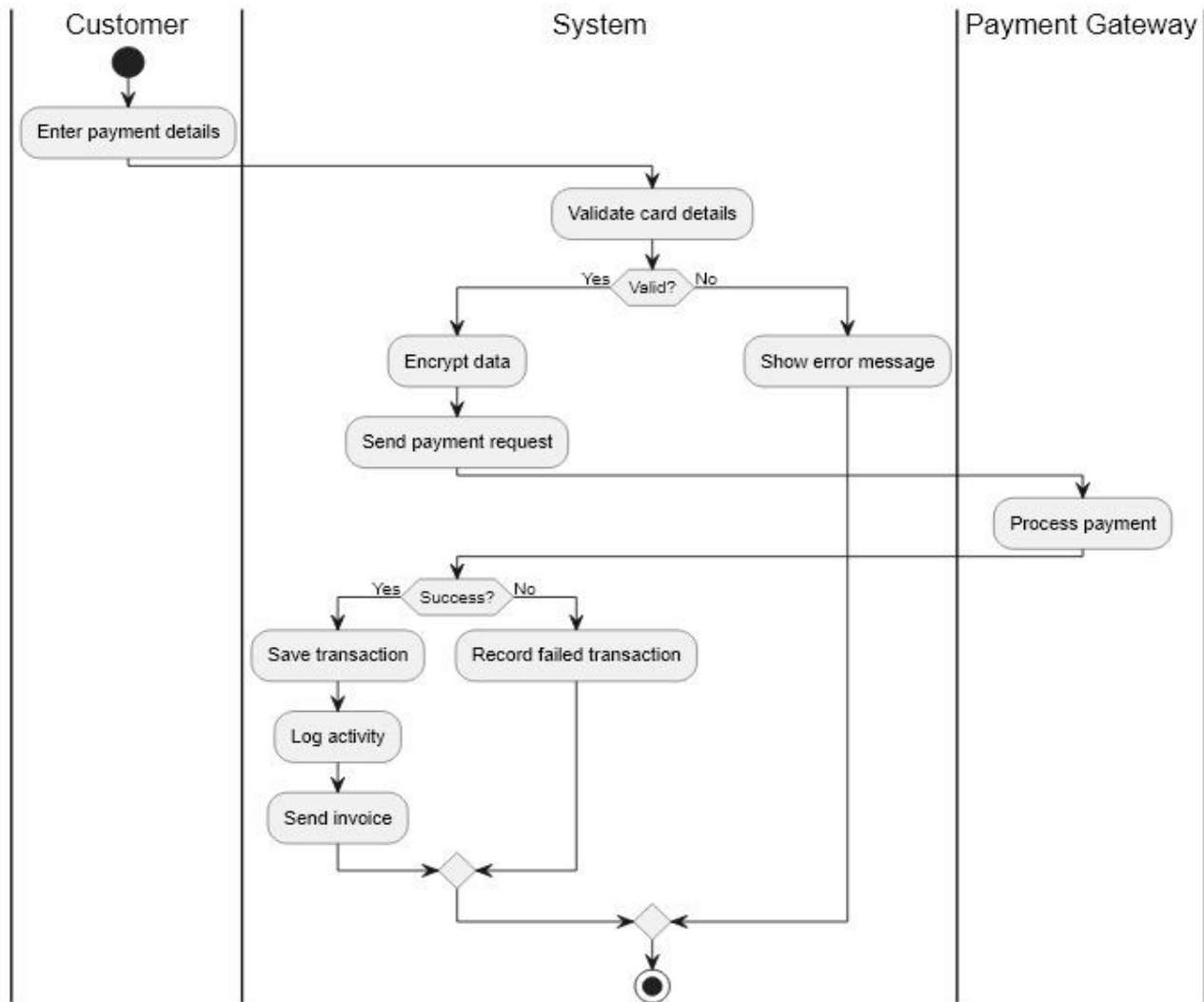


Figure 13: BPMN diagram – payment management

This BPMN diagram represents the payment processing workflow involving the Customer, System, and Payment Gateway. The system validates card details and sends an encrypted request to the payment gateway for transaction processing. If the payment is successful, the transaction is stored, activity is logged, and an invoice is generated. If the payment fails, the system records the failed transaction and displays an error message to the customer.

3) Order management

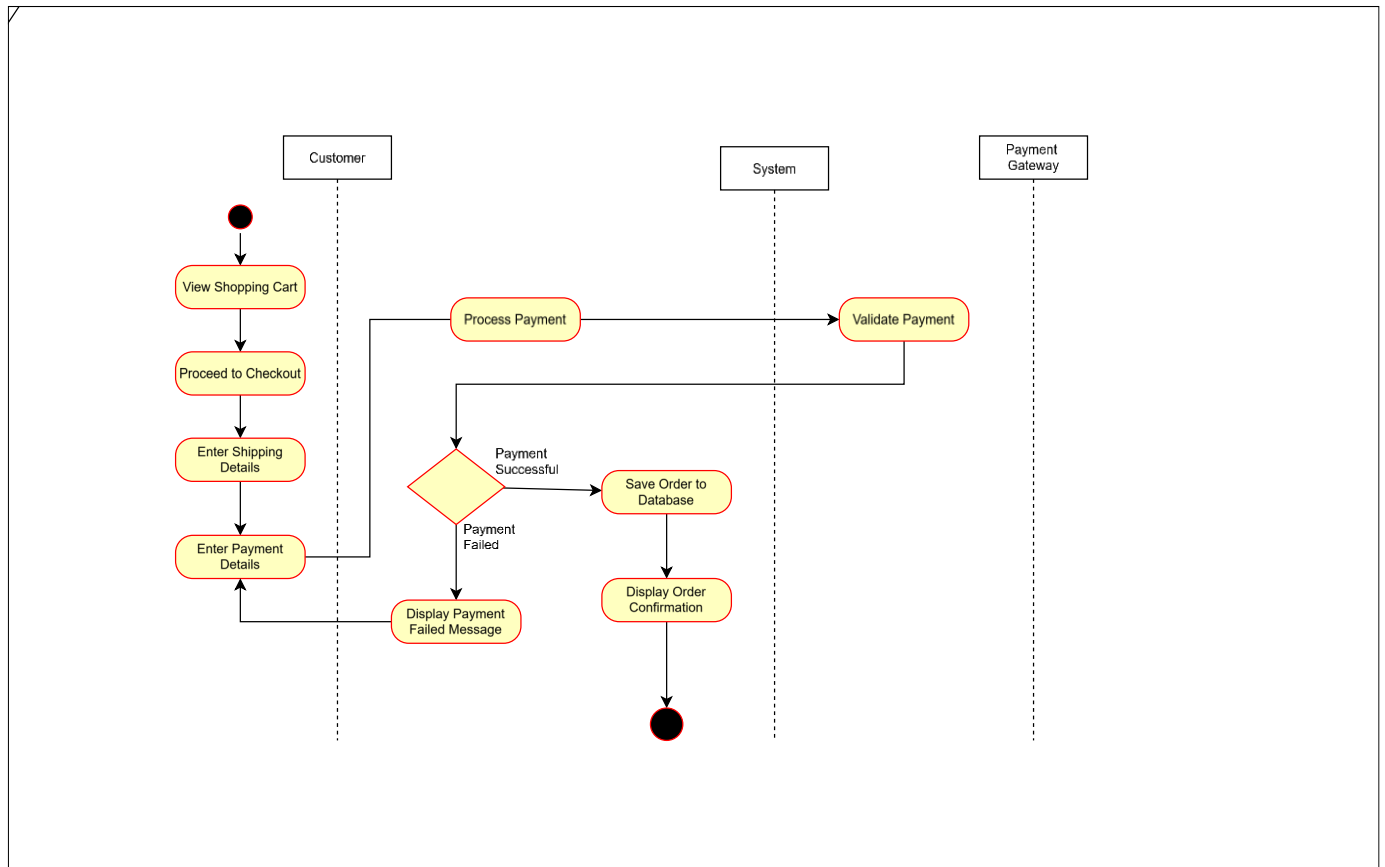


Figure 14: BPMN diagram –order management

This BPMN diagram illustrates the order checkout workflow performed by the Customer and the System during order placement. It includes steps such as viewing the shopping cart, entering shipping and payment details, and validating payment through the payment gateway. Based on the payment result, the system either stores the order and confirms success or displays a failure message. This process ensures accurate order processing and confirmation.

4) Delivery management

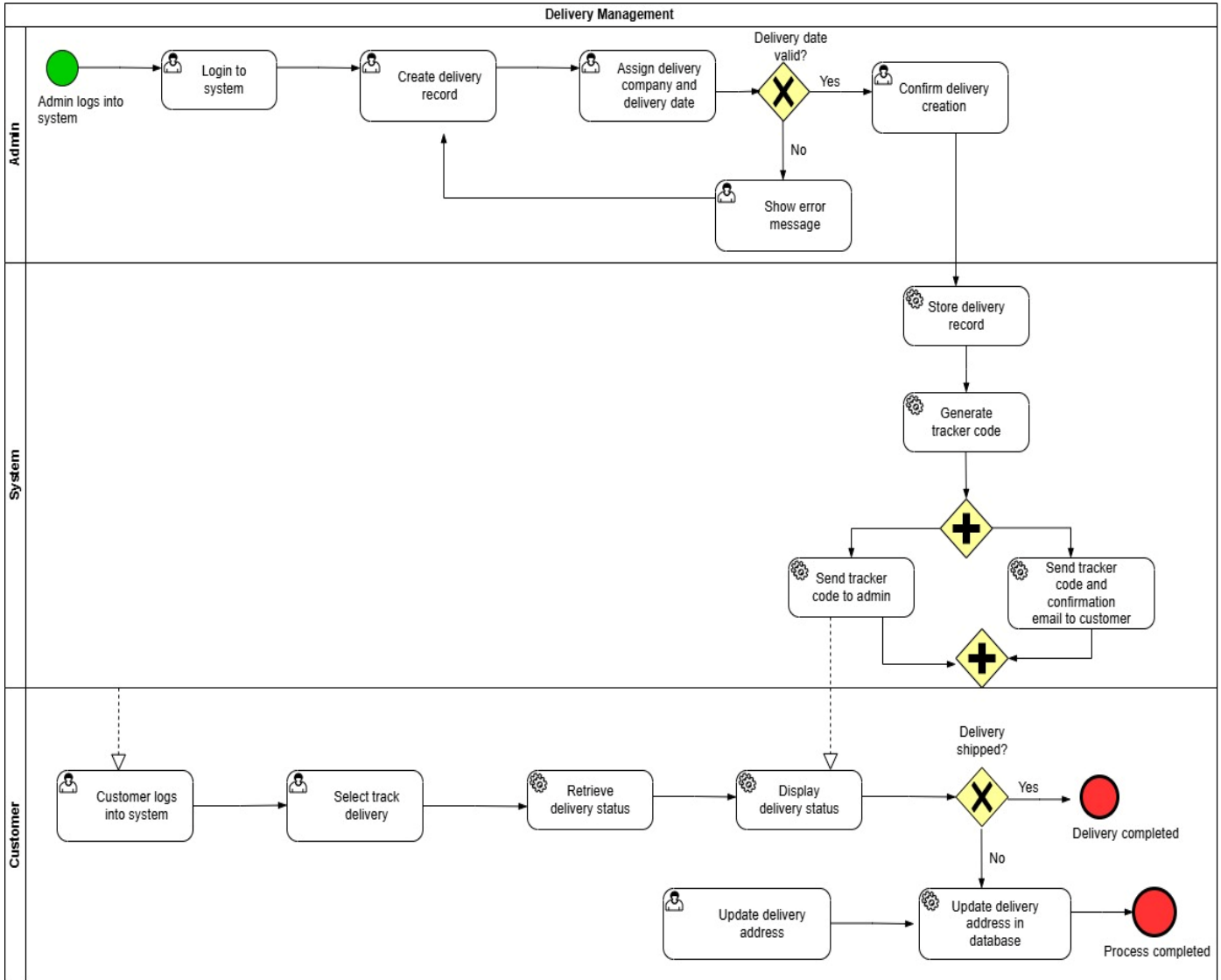


Figure 15: BPMN diagram –Delivery management

This BPMN diagram illustrates the delivery management workflow involving the Delivery Manager, System, and Customer. It shows how delivery records are created, validated, stored, and tracking codes are generated and shared with both the admin and the customer. The customer can log into the system to monitor delivery status and update the delivery address if the shipment has not yet been completed. This process supports efficient coordination and tracking of delivery operations within the system.

DECISION TREES

1) Payment Management

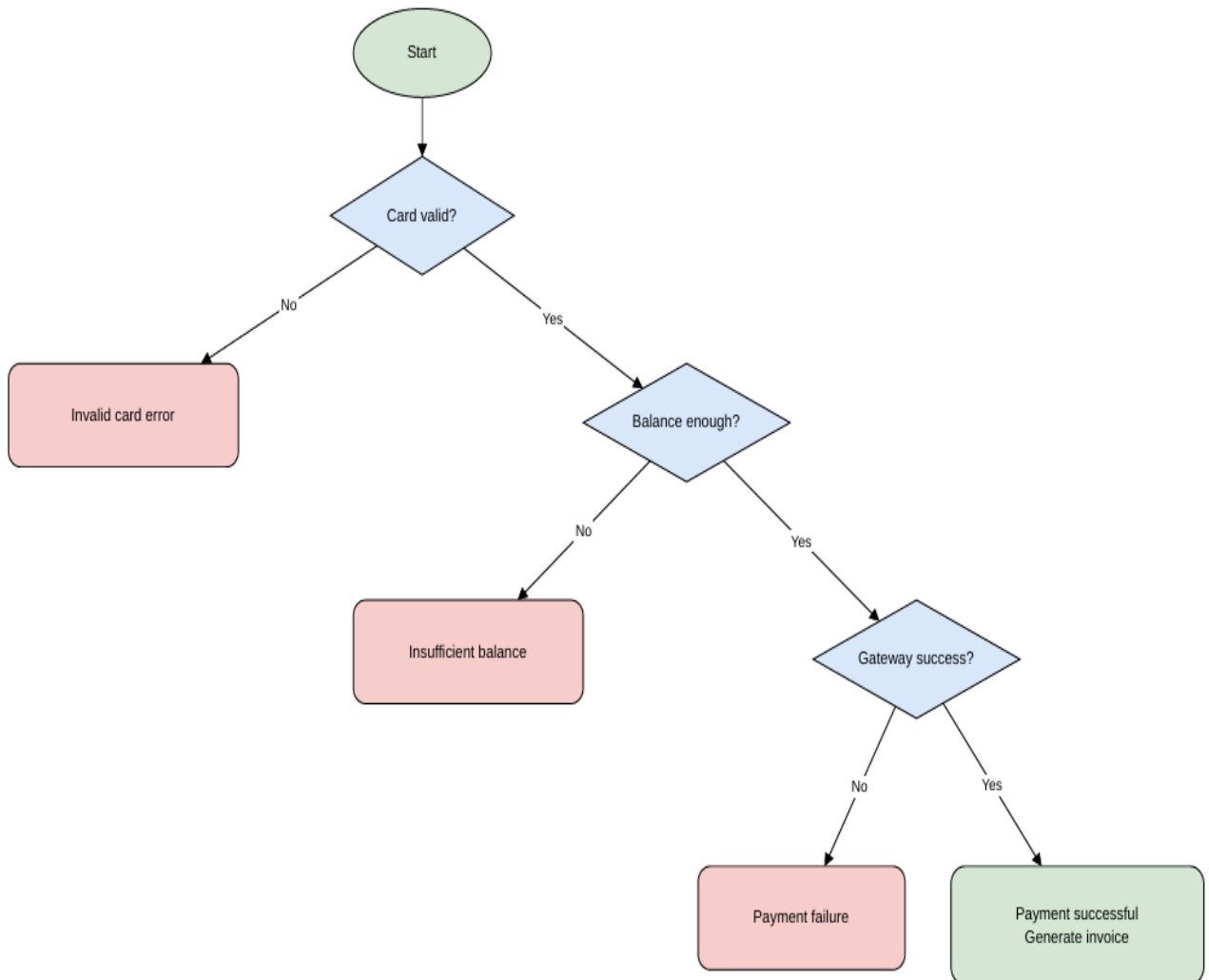
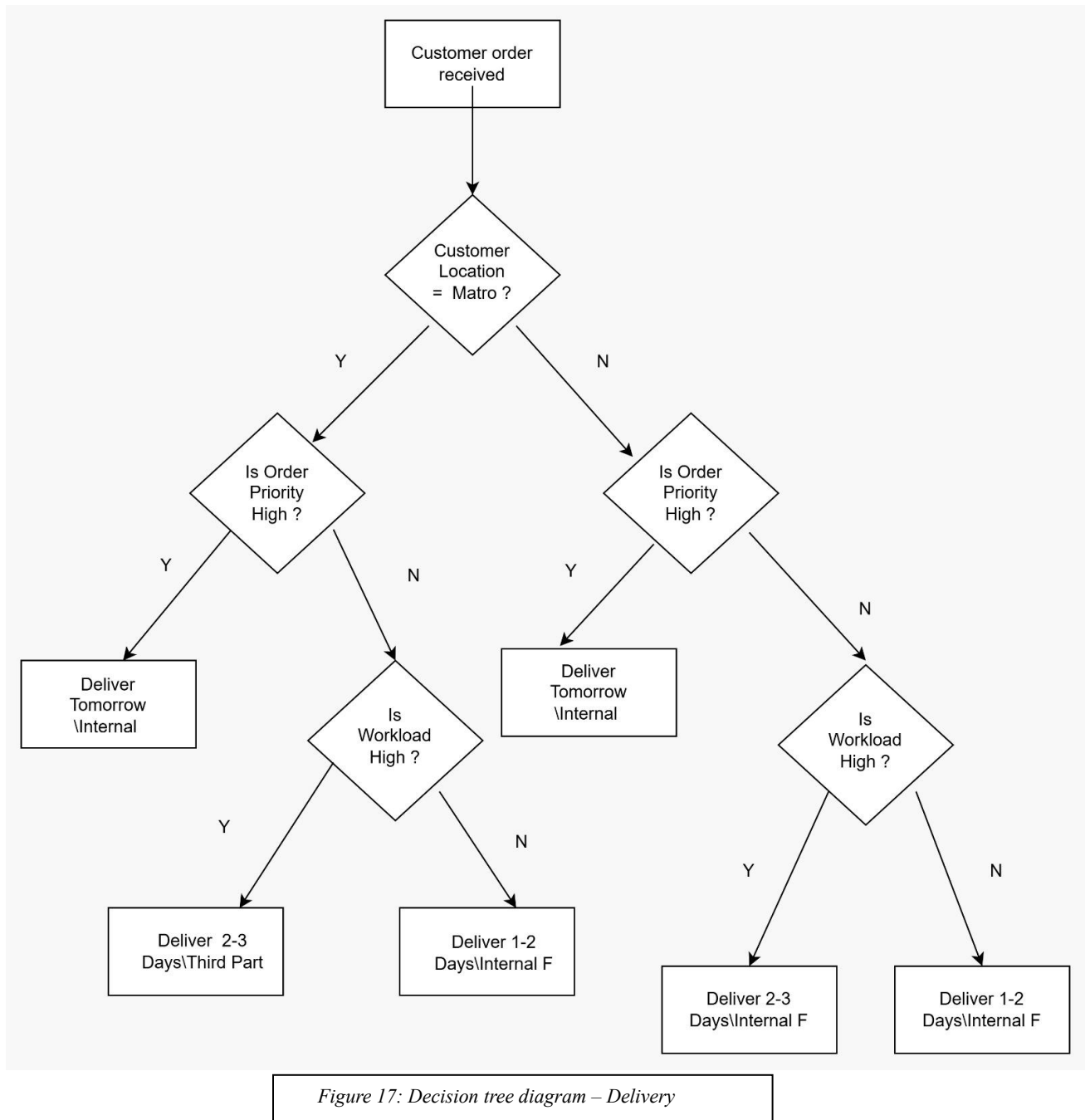


Figure 16: Decision tree diagram – payment

This decision tree illustrates the decision logic used during payment processing in the system. It evaluates whether the card details are valid, whether sufficient account balance is available, and whether the payment gateway transaction is successful. Based on these conditions, the system either generates an invoice for successful payments or displays appropriate error messages. This model helps represent the conditional flow of payment validation clearly.

2) Delivery management



This decision tree represents the decision logic used to determine delivery scheduling based on customer location, order priority level, and current workload conditions. Depending on these factors, the system selects the appropriate delivery method and estimated delivery timeframe. The model helps support efficient delivery planning by guiding internal or third-party delivery assignment decisions. This improves coordination of delivery operations within the system.

DECISION TABLE

1) Payment management

Decision Table – Payment System

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Card valid?	Yes	Yes	No	Yes
Balance enough?	Yes	No	-	Yes
Payment gateway success?	Yes	Yes	-	No

Actions	Rule 1	Rule 2	Rule 3	Rule 4
Payment successful	Yes	No	No	No
Show insufficient balance	No	Yes	No	No
Show invalid card error	No	No	Yes	No
Show payment failure	No	No	No	Yes
Generate invoice	Yes	No	No	No

Figure 18: Decision table diagram – payment management

This decision table represents the conditional logic used in the payment processing workflow of the system. It evaluates card validity, account balance availability, and payment gateway response to determine the appropriate system action. Based on these conditions, the system either confirms payment success and generates an invoice or displays relevant error messages. This table ensures clear representation of payment-related decision rules.

2) Delivery management

Conditions	R1	R2	R3	R4	R5	R6	R7	R8
Customer Location = Metro?	Y	Y	Y	Y	N	N	N	N
Order Priority = High?	Y	Y	N	N	Y	Y	N	N
Current Workload = High?	-	-	Y	N	-	-	Y	N
Actions								
Delivery Date = Tomorrow	X				X			
Delivery Date = 1-2 Days		X		X		X		X
Delivery Date = 2-3 Days			X				X	
Assign to Internal Fleet	X	X		X	X	X	X	X
Assign to Third Party			X					

Figure 19: Decision table diagram –delivery management

This decision table illustrates the logic used to determine delivery scheduling based on customer location, order priority level, and current workload conditions. Depending on these conditions, the system selects an appropriate delivery timeframe and assigns either internal delivery resources or third-party services. The table supports structured decision-making for efficient delivery planning and resource allocation.

WIREFRAMES

1) Login / Registration Screen - IT24100696 – Dissanayake L.N

Create Account

First Name *

Last Name *

Email *

Password *

(min 8 characters)

Phone(optional)

Preferred Language * ▼

English

Create Account

Don't have an account? Sign up

• Please fill all required fields

Figure 20.1: Registration screen

This wireframe illustrates the user registration interface used by new customers to create an account in the Integrated Furniture Shop Management System. It allows users to enter personal details such as name, email, password, phone number, and preferred language before submitting the registration request. The interface supports account creation and validation processes within the User Account Management module.

Login

Email *

Password *

Login

Forgot password?

Don't have an account? Sign up

Figure 20.2: Login screen wireframe

This wireframe represents the login interface used by registered users to access the system securely. It allows users to enter their email and password credentials and provides options for password recovery if needed. This interface supports secure authentication within the User Account Management module.

2) Product List screen

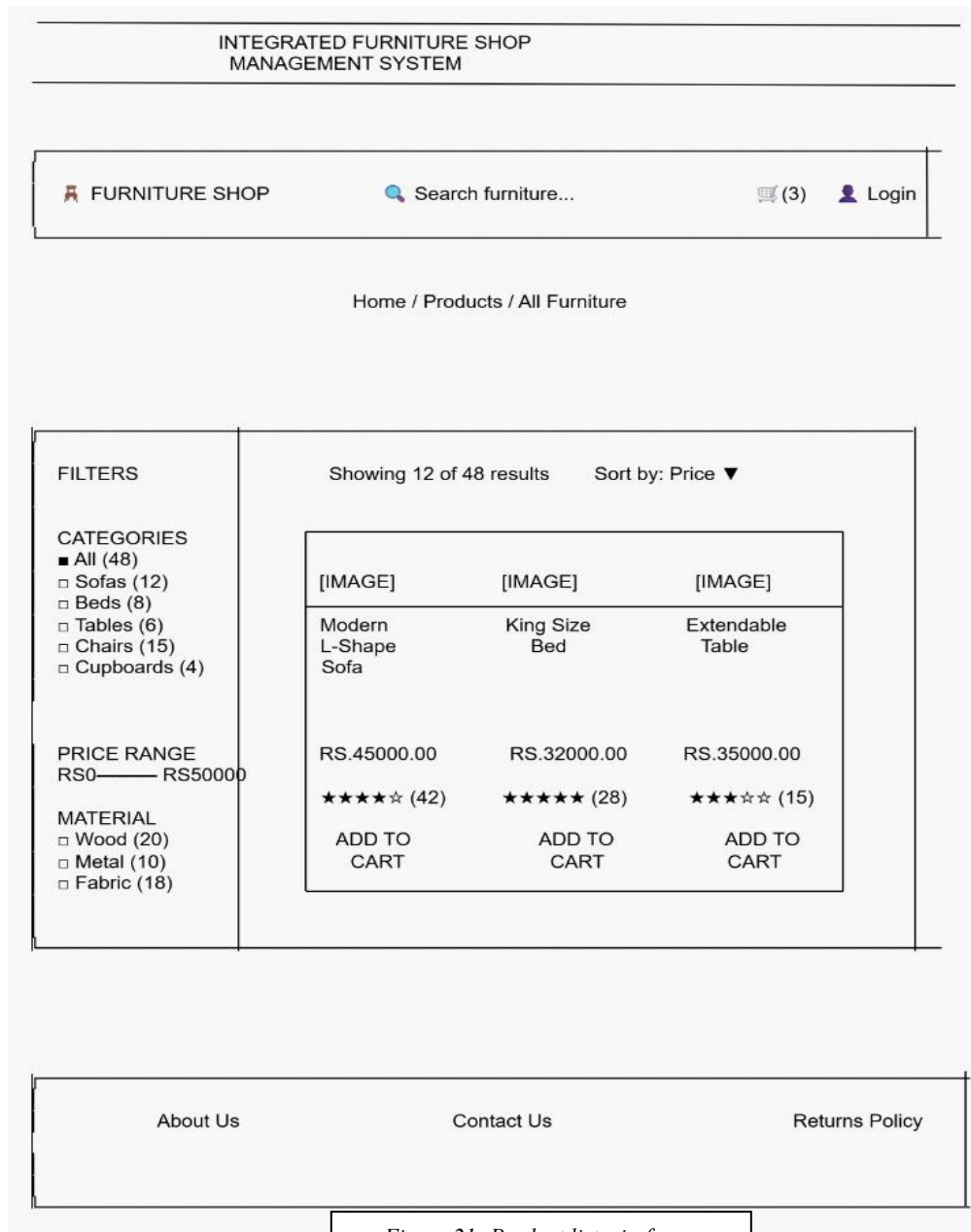


Figure 21: Product list wireframe

This wireframe illustrates the product browsing interface of the Integrated Furniture Shop Management System, where customers can search, filter, and view available furniture items by category, price range, and material type. It allows users to review product details such as name, price, and ratings before adding items to the shopping cart. This interface supports product selection and ordering functions within the Product Management and Order Management modules.

3) Payment screen

PAYMENT METHODS

☒ VISA/MASTER CARD ☐ GOOGLE PAY ☐ KOKO

Use Saved Cards

[Dropdown Menu]

Card Holder Name

Card Name

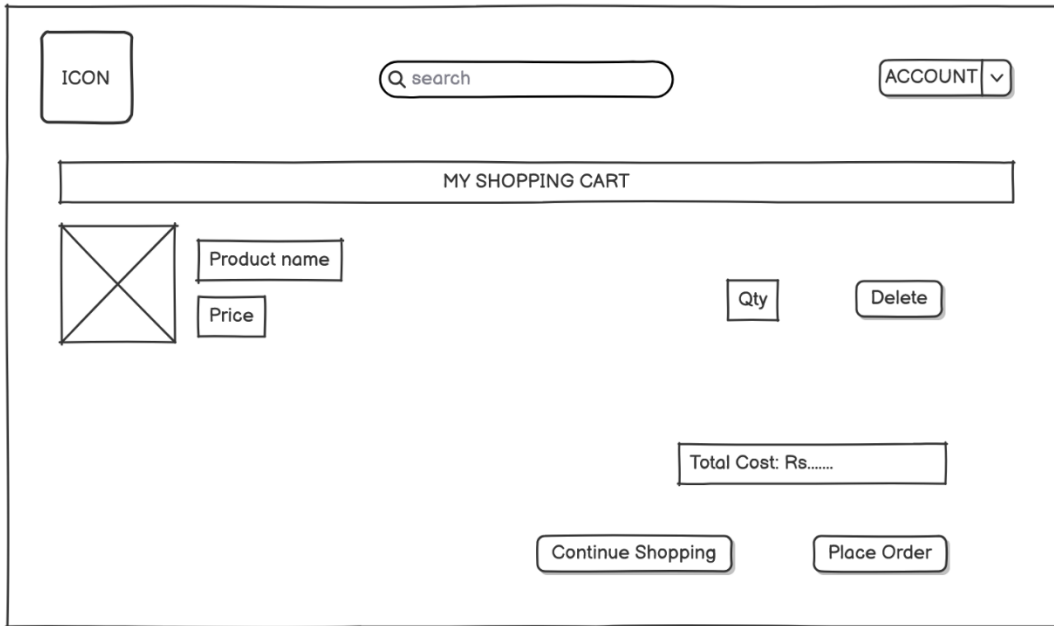
MM/YY CVV

☐ Save Card Details

Figure 22: payment wireframe

This wireframe illustrates the payment interface used by customers to complete transactions in the Integrated Furniture Shop Management System. It allows users to select a preferred payment method such as Visa/MasterCard, Google Pay, or KOKO, and enter card details securely. The interface also supports selecting saved cards and saving new card information for future use. This screen supports the Payment and Billing module of the system.

4) Shopping cart screen

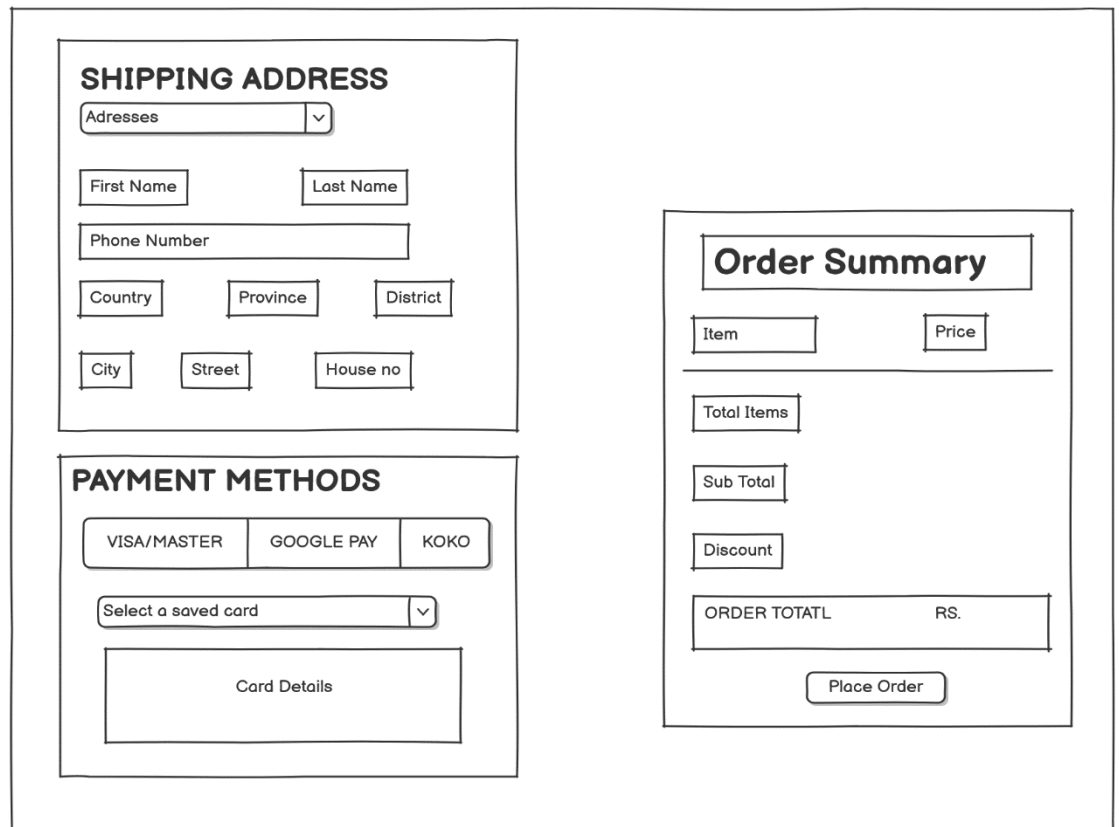


The wireframe for the shopping cart screen includes a top navigation bar with an 'ICON' placeholder, a search bar with a magnifying glass icon and the text 'search', and an 'ACCOUNT' dropdown menu. Below this is a section titled 'MY SHOPPING CART'. The main area contains a table with columns for a product image (represented by a box with an 'X'), 'Product name', 'Price', 'Qty', and 'Delete'. At the bottom of the cart area, there is a 'Total Cost: Rs.....' label and two buttons: 'Continue Shopping' and 'Place Order'.

This wireframe illustrates the shopping cart interface where customers can review selected products before placing an order. It displays product details such as item name, quantity, price, and total cost, and allows users to remove items or continue shopping. The interface supports order preparation within the Shopping Cart and Order Management module.

Figure 23.1: shopping cart wireframe

This wireframe illustrates the checkout interface where customers enter shipping address details, select payment methods, and review the order summary before confirming the purchase. It provides a structured view of order items, subtotal, discounts, and total payment amount. This interface supports the order confirmation process within the Order Management and Payment modules.



The checkout interface is divided into three main sections. The 'SHIPPING ADDRESS' section includes a dropdown for 'Adresses', and input fields for 'First Name', 'Last Name', 'Phone Number', 'Country', 'Province', 'District', 'City', 'Street', and 'House no'. The 'PAYMENT METHODS' section features buttons for 'VISA/MASTER', 'GOOGLE PAY', and 'KOKO', a dropdown for 'Select a saved card', and a 'Card Details' input area. The 'Order Summary' section on the right lists 'Item' and 'Price' columns, followed by 'Total Items', 'Sub Total', and 'Discount'. At the bottom, it shows 'ORDER TOTATL' and 'RS.' with a 'Place Order' button.

Figure 23.2: shopping cart wireframe

5) Delivery tracking screen

AunaraFurnitures

← → ↻ <https://AnuraFurnitures.com/delivery-tracking>

AnuraFurnitures

Track Your Delivery

Enter your tracking code to view delivery status

Order Details

Order ID: #3

Status: Pending

Delivery Partner: PTD Courier

Delivery Progress

Confirmed Shipped On the Way Delivered

Estimated Delivery: 10/04/2026

Figure 24.1: Delivery tracking screen wireframe

This wireframe illustrates the delivery tracking interface used by customers to monitor the status of their orders by entering a tracking code. It displays key delivery information such as order ID, delivery partner, current delivery status, delivery progress stages, and estimated delivery date. The interface enables customers to track shipment progress from confirmation to final delivery. This screen supports delivery monitoring within the Delivery Management module of the system.

AunaraFurnitures

← → ↻ https://AunaraFurnitures.com/delivery-tracking

Track Your Delivery

Enter your tracking code to view delivery status

Figure 24.2: Delivery tracking screen wireframe

This wireframe illustrates the delivery tracking code entry interface used by customers to access delivery status information. It allows users to enter a tracking code and request real-time delivery progress details through the system. The interface serves as the entry point for monitoring shipment updates. This screen supports delivery tracking functionality within the Delivery Management module of the system.

6. Support Ticket Screen

Anura Furnitures

Login / Profile

Shop My Tickets Track Ticket

My Support Tickets

Create New Ticket

Contact Number

Request Type

Complaint

Subject

Message

Attachments

Figure 25: Support ticket wireframe

This wireframe illustrates the support ticket submission interface used by customers to report issues, complaints, or service requests to the system. It allows users to enter contact details, select request type, provide a subject and message description, and attach supporting files if necessary. The interface supports communication between customers and the support team. This screen supports the Feedback and Support module of the system.

REFERENCES LIST

1. Draw.io Diagram Tool -<https://draw.io/>
2. Balsamiq Diagram Tool - <https://balsamiq.com/>
3. IE2021 – Requirements Engineering (2026) Assignment 2: Requirements Specification -
https://courseweb.sliit.lk/pluginfile.php/538924/mod_resource/content/1/RE%20Assignment%202.pdf
4. Requirement Engineering Modeling Behavior -
https://courseweb.sliit.lk/pluginfile.php/535262/mod_resource/content/1/Modeling%20Behavior%20-6%20slides.pdf
5. Requirement Engineering Modeling Structure -
https://courseweb.sliit.lk/pluginfile.php/543225/mod_resource/content/1/Modeling%20Structure%20-%20%20%206%20slides.pdf
6. Requirement Engineering Decision Tables and Trees -
https://courseweb.sliit.lk/pluginfile.php/546162/mod_resource/content/1/4%20DecisionT_6SLIDES.pdf
7. Requirement Engineering Lab Sheet 8 - Topic: Functional & Process Modeling –
https://courseweb.sliit.lk/pluginfile.php/551007/mod_resource/content/0/IE2021-LabSheet-08-2026-S2.pdf

GROUP MEMBERS CONTRIBUTION TABLE

Group Member	Contribution	Pages
Perera A.J.M.P	UML Sequence Diagrams – Payment Management , UML Activity Diagrams – Payment Management , BPMN Diagram – Payment Management , Wireframe: Payment Screen	7,32,36,41,49
Dissanayake L.N	UML Class Diagram, UML Sequence Diagram – User Registration, UML Activity Diagrams – User Registration, BPMN Diagram – User Registration, Wireframe - Login / Registration Screen	3,4,29,31,35,40,47,
Sewwandi R.P.D.D.C	Decision Table – Delivery Management , Decision Tree – Delivery Management, UML State Diagram– Order, Wireframe - Product List Page	26,39,45,46,48
Wickramarachchi W.A.M.V	Decision Table – Payment Management , Decision Tree – Payment Management, UML Physical Diagram, Wireframe: Support Ticket Screen	27,30, 44,46,52
Bandara N.H.M.W.W.D.S	UML Sequence Diagram – Delivery Management , UML Activity Diagrams – Delivery Management, BPMN Diagram – Delivery Management, Wireframe: Delivery Tracking Screen Prepared and compiled the complete Document	34,38,43,50,52, 53,54,55
Ananda U.V.D.M.A	UML Sequence Diagram – Order Management , UML Activity Diagrams – Order Management, BPMN Diagram – Order Management, Wireframe: Shopping Cart Screen	33,37,42,50,26,56

APPENDICES

Appendix A — Assumptions

The following assumptions were made during the requirements analysis and modelling process:

- The furniture shop operates as a medium-scale retail business.
- Customers access the system through a web-based interface.
- Online payments are processed using third-party payment gateway services.
- Delivery tracking information is updated by the delivery manager based on courier updates.
- Internet connectivity is available for all system users.
- Appendix B — Modelling Tools Used

Appendix B — Modelling Tools Used

The following tools were used to create the system models and diagrams:

- Draw.io — Activity diagrams, BPMN diagrams, decision trees, and decision tables
- Balsamiq — Wireframe mock-up interfaces
- Microsoft Word — Documentation preparation